

4. A group of students investigate density using an electronic balance, a measuring cylinder, water and some small steel ball bearings.

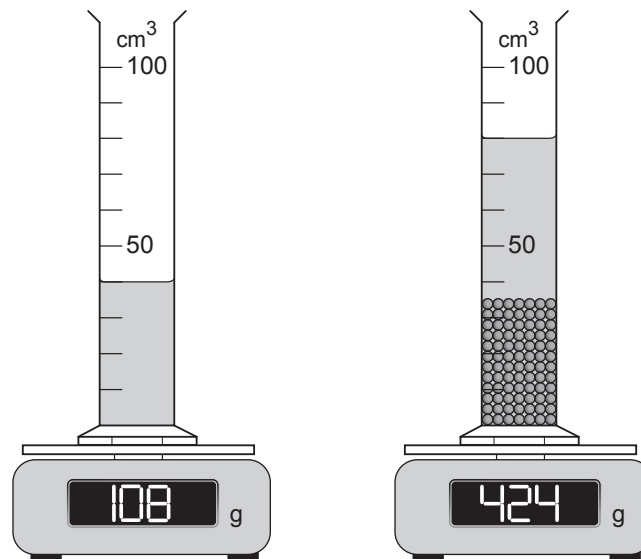
They set the balance to zero and place a measuring cylinder on to the balance.

They then add water at 10 cm^3 intervals and record both the mass and volume.

At some point they start adding the steel ball bearings instead of water, and record the mass and volume each time the water level rises by 10 cm^3 .

They continue until the final volume of 100 cm^3 is reached.

The arrangement is shown below.



The students produce a table of their results, which is shown below.

Volume (cm^3)	Mass (g)
0	68
10	78
20	88
30	98
40	108
50	187
60	266
70	345
80	424
90	503
100	582



- (a) One member of the group claims that water has a density value of 1 g/cm^3 . Explain whether her claim is correct. [No calculations are required.] [1]

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- (b) (i) State the volume at which they start adding the steel ball bearings. [1]

Volume = cm^3

- (ii) Give a reason for your answer. [1]

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- (c) Use an equation from page 2 to calculate the density of steel. [2]

Density of steel = g/cm^3

- (d) State **two** improvements that could be made to this experiment that would lead to a more accurate value for the density of steel. [2]

1.

2.

